Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	26 February
Team ID	
Project Name	**An Exploration of Electricity Consumption Patterns**
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Acquisition & Integration	Import state-wise electricity dataset (Jan 2019– Dec 2020) Integrate CSV, SQL, and Cloud data sources. Validate dataset completeness and consistency. Support external API integration (optional weather data)
FR-2	Data Cleaning & Preprocessing	Remove duplicate records Handle missing values using statistical methods Convert date column into time-series format Standardize state and regional classifications Detect and treat outliers (IQR / Z-score)
FR-3	Overall Consumption Trend Analysis (Scenario 1)	Calculate national monthly electricity consumption Compare 2019 vs 2020 consumption trends Analyze lockdown impact (March–June 2020) Compute percentage growth/decline Generate time-series visualizations

FR-4	Regional Demand Analysis (Scenario 2)	Group states into regions (North, South, East, West, Northeast) Calculate region-wise monthly totals Compare regional growth patterns Identify highest and lowest demand regions Generate comparative charts and heatmaps
FR-5	Recovery Analysis PostLockdown (Scenario 3)	Compare pre-lockdown, lockdown, and post-lockdown periods Calculate recovery percentage per state Identify states exceeding pre-COVID levels Perform recovery trend visualization Optional: Apply ML forecasting model
FR-6	Dashboard & Reporting	Create interactive dashboard (Power BI/Tableau/Streamlit) Enable filters (state, region, month, year) Export reports (PDF/Excel) Display key insights & summary statistics

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	User-friendly interface with easy navigation and clear visualizations
NFR-2	**Security**	Secure authentication, role-based access, encrypted credentials
NFR-3	**Reliability**	Accurate statistical computations and consistent results
NFR-4	**Performance**	Efficient handling of large datasets and optimized queries

NFR-5	**Availability**	24/7 system availability with cloud deployment option
NFR-6	**Scalability**	Support future data expansion and large-scale processing